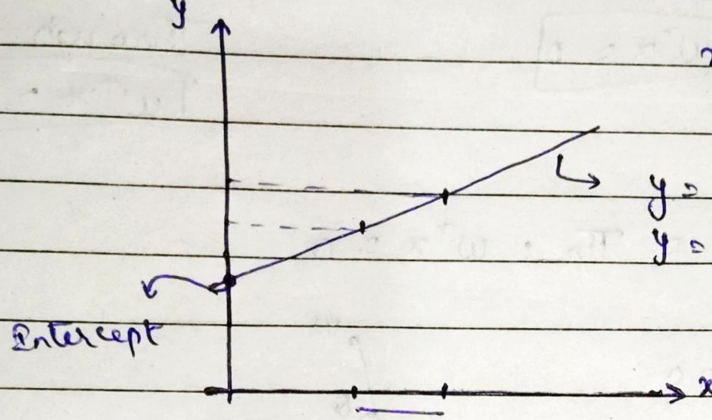
* Equation of Line, 3d plane and Hyperplane (n Dimension).



$x > 0$

$y = mx + c$ → m = Slope, c = Intercept

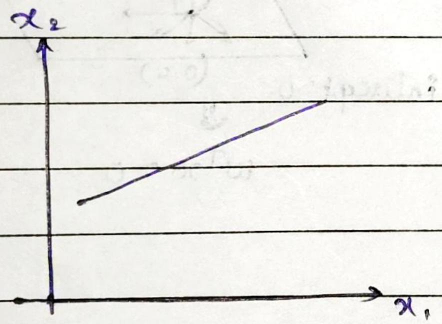
$y = \beta_0 + \beta_1 x$

$ax + by + c = 0$

$by + c = -ax$

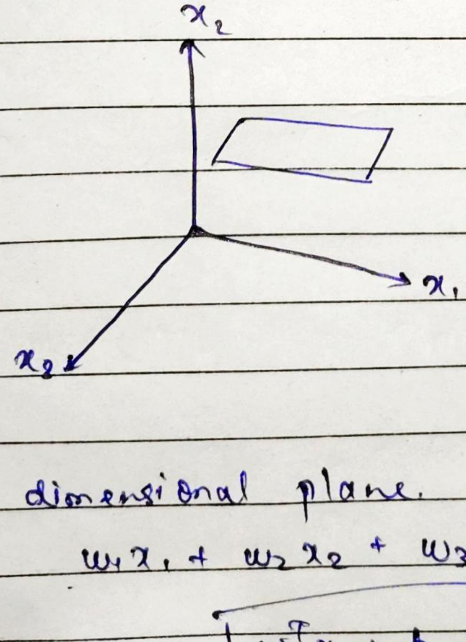
$by = -ax - c$

$y = mx + c \Leftarrow y = \underbrace{\boxed{-\frac{a}{b}}}_{m} x - \underbrace{\boxed{\frac{c}{b}}}_{c}$



$w_1 x_1 + w_2 x_2 + b = 0$

$\boxed{w^T x + b = 0}$ ⇒ Eqn of straight line.



$w_1 x_1 + w_2 x_2 + w_3 x_3 + b = 0$

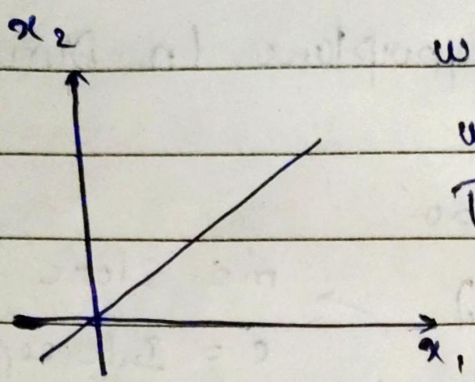
$\boxed{w^T x + b = 0}$

$w = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

→ n-dimensional plane.

$w_1 x_1 + w_2 x_2 + w_3 x_3 + \dots + w_n x_n + b = 0$

$\boxed{w^T x + b = 0}$



$w_1 x_1 + w_2 x_2 + b = 0$

$w_1 x_1 + w_2 x_2 = 0$ ($\because b = 0$)

$\boxed{w^T x = 0}$

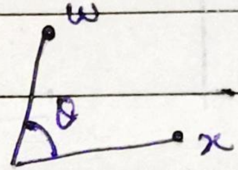
• Equation of a St. line passing through an origin

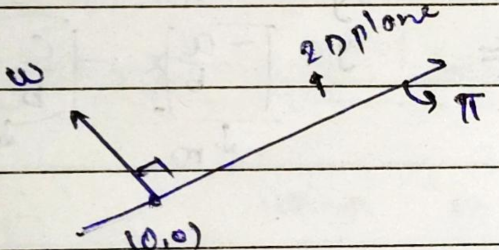
$\boxed{w^T x = 0}$

— Equation of plane = $\pi_n : w^T x = 0$

$w^T x = 0$

$w \cdot x = w^T x = \|w\| \, \|x\| \cos\theta = 0$

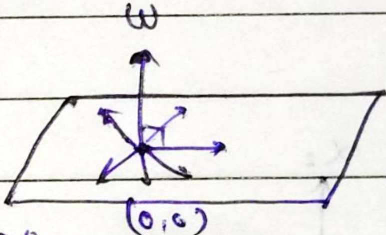




$\theta = 90°$

$\cos\theta = 0$

intercept = 0.



$\Downarrow$

$w^T x = 0$

$\boxed{w \perp^r \pi}$